PA194N
A stack-preserving modus-ponens machine for the consistency of Peano arithmetic

Comparative research and executable verification

13 November 2026, Pacific time

Abstract

We compare the 41 research packets represented in the requested repository snapshot and construct a new deterministic, binary, one-tape, blank-start machine with **194 working states**. We do not obtain a verified machine with fewer than 194 states. The accepted result is the requested alternative: PA194N changes the successful modus-ponens command so that it preserves the proof stack, while retaining the original Shift test $t + h = 1$. Its exact transition graph is absent from all eight previously represented 194-state classes, including supplementary and rejected tables.

The construction retains the primary 67-reader, seven-fold axiom decoder. One native transition bypasses a 19-instruction dummy-zero push. The resulting source has 3,654 primitive sites and 4,755 symbolic outcomes. Both physical checkers pass every outcome for arbitrary natural register values and an arbitrary nonempty spare list; an independent checker rederives the generic register-loop lemmas from the actual rows. Exact blank initialization takes 58,154 transitions and is checked against the complete tape.

A ten-entry command compiler proves completeness of the changed interface. In particular, when the offset is zero, $\mathsf{MP}(0)$ alone resets an arbitrary current formula without changing the stack. Together with syntactic proof annotation for soundness, positive physical simulation, and the explicitly inherited PA calibration of the separate-diagonal-head calculus, the ordinary mathematical argument gives

$$\mathrm{PA} \vdash \big(\neg\,\mathrm{Halt}(M_{194N}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})\big).$$

This is not a newly kernel-checked Lean or Rocq endpoint. The archive supplies the literal table, deterministic construction, independent replay, mathematical proofs, complete corpus census, search records, and all LaTeX sources.

Contents

1 Outcome, conventions, and exact identity **3**

2 The substantive change: do not push a dummy zero **3**
2.1 One literal row and nineteen primitive sites . . . 4
2.2 A simpler exact reset . . . 4

3 A complete ten-entry command compiler **5**
3.1 Exact numerical operations and command contracts . . . 5
3.2 Uniform stack operations and a one-command discard . . . 6
3.3 Axiom selectors and generalization . . . 6
3.4 Modus ponens with its spare zero supplied by the compiler . . . 7
3.5 Soundness and the PA calibration . . . 7

4 The inherited logical calibration and PA internalization **8**

5 Physical construction and finite-trace preservation **9**

6 Verification results and exact scope **10**

7 Comparison with the complete supplied corpus **11**
7.1 The eight existing 194-state graphs . . . 11
7.2 The positive 195-state families and their limitations . . . 12
7.3 What the larger designs contribute . . . 13
7.4 Proof status and inherited assumptions . . . 13
7.5 Coverage limits and reproducibility . . . 14
7.6 Prior work on expanded macro searches . . . 14

8 Search below 194 and its limits **14**
8.1 Deeper macro expansion and broader fold allocation . . . 15
8.2 Additional physical contraction searches . . . 15
8.3 What transfers from ZFC and concise iPRA . . . 16

9 Reproduction and package organization **16**

A Complete transition table **17**

B Archive-by-archive ledger **18**

C Independent review and artifact boundary **20**

Sources and attribution **21**

1 Outcome, conventions, and exact identity

The primary deliverable is `construction/machine/pa194N.tm`. The machine has working states $0, \ldots, 193$, initial state 0, and a separate terminal label 194. The head starts at coordinate zero of a two-way infinite all-zero tape. The alphabet is $\{0, 1\}$; every transition writes one bit and moves one cell left or right. There are 388 rows and no row for the halt label. No input, external initialization, stationary move, or change of state-counting convention is used.

The exact dense table SHA-256 is

`37f7e909724cab7fd667db28f89b9c81653d2576c033750e783b40b3289c04f9`.

The native presentation starts in 74 and halts in 254. Its SHA-256 is

`b38dfe8cbd6e84aadbea5912e20f25b0bbda8f3fad4fbec0c779a77c857a8dc8`.

A supplied bijection checks every native/dense transition. All 194 working states are reachable in the designated finite transition graph. This does not assert that every state occurs on the actual blank run.

Machine	Working states	Primitive source sites	Primitive outcomes
Primary PA194	194	3,673	4,780
PA194C	194	3,649	4,749
PA194CS	194	3,646	4,745
PA194N	**194**	**3,654**	**4,755**

PA194N offers a different command protocol; it is not a claim to have the shortest source among the 194-state machines. The search did not establish 193 states or a lower bound excluding them.

Theorem 1.1 (Ordinary PA calibration of the delivered table)**.** *Let H_N be the arithmetical assertion of a finite halting computation of the specified literal machine from its blank start. Using the published syntactic PA calibration of the retained numerical calculus,*

$$\mathrm{PA} \vdash H_N \leftrightarrow \exists p \, \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner), \qquad \mathrm{PA} \vdash \neg H_N \leftrightarrow \mathrm{Con}(\mathrm{PA}).$$

The proof has three distinct interfaces: the numerical-calculus/PA proof translations, the new proof-to-command compiler and annotated command execution, and the literal source/tape simulation. The latter two are specified below for PA194N. The inherited logical theorem is described explicitly in Section 4; a separate conditional iPRA assembly in the repository is not used as an unconditional proof of this theorem. Actual nonhalting follows under the assumption that PA is consistent.

2 The substantive change: do not push a dummy zero

Write a command store as (t, w, d, h): the current formula, a numerical Cantor stack, an offset, and the cached formula. Put

$$P(a, b) = \frac{(a + b)(a + b + 1)}{2} + b, \quad [a; w] = P(a, w), \quad I(a, b) = 4P(a, b).$$

The ordinary zero-stack convention is literal: $[0; 0] = 0$. Formula code zero denotes truth and code one denotes the designated contradiction. For a modus-ponens parameter x, set $b = d + x$. The successful comparison is $I(t, b) = h$. The three relevant interfaces are

Controller	Successful $\mathsf{MP}(x)$ result	Shift halt test
Original PA194	$(b, [0; w], b, 0)$	$t + h = 1$
Cached PA194C	$(0, w, 0, b)$	$h = 1$
PA194N	$\boldsymbol{(b, w, b, 0)}$	$\boldsymbol{t + h = 1}$

The failed comparison in PA194N returns $(0, w, b, 0)$. Its nonhalting Shift is the original

$$\mathsf{S}(t, [a; v], d, h) = (h, v, d, a), \qquad \text{provided } t + h \neq 1.$$

Thus PA194N keeps the consequent in the current formula and offset registers, preserves the stack on both MP branches, and retains the original two-formula halt guard. It is neither the original protocol nor the cached protocol with controls renamed.

For example, take $(t, w, d, h) = (2, 5, 0, 72)$ and parameter $x = 3$. Since $I(2, 3) = 72$ and $[0; 5] = 20$, the original controller returns $(3, 20, 3, 0)$, PA194N returns $(3, 5, 3, 0)$, and the cached controller returns $(0, 5, 0, 3)$. This is a direct operational distinction, not merely a difference between table hashes.

2.1 One literal row and nineteen primitive sites

In the primary PA194 native table make exactly this replacement:

$$\delta(164, 1) = (1, R, 98) \quad \longmapsto \quad (1, R, 114).$$

The old target implements `node55=(node49,node54)`. The new target starts directly at `node54`. Here `node49` is the Cantor-stack push and `node54` is the preserving copy of D into T through scratch register X. After successful comparison, $T = H = X = 0$ and $D = b$. The skipped push would therefore have pushed zero. The retained loop returns $T = b, D = b, X = 0$, giving the stated interface.

The retained copy loop has an elementary all-natural proof: after $j \leq b$ iterations of its first phase, $(T, D, X) = (j, b - j, j)$; after $j \leq b$ iterations of its restoring phase, $(T, D, X) = (b, j, b - j)$. It exits with $(b, b, 0)$ after exactly $7b + 2$ primitive source steps.

The source splice removes old primitive ordinals $[3634, 3653)$, a block of 19 sites. Its only external entry is the edge $3628 \to 3634$. The builder checks every surviving instruction and both targets of each decrement under the explicit ordinal relocation. The seven-instruction remaining copy loop is written out in the construction receipt. This binds the mathematical command change to the actual source, instead of inferring it from a sampled execution.

2.2 A simpler exact reset

A useful consequence of the changed stack behavior is the stronger identity

$$\mathsf{MP}(0)(t, w, 0, h) = (0, w, 0, 0) \qquad (t, w, h \in \mathbb{N}).$$

Here the output parameter $b = d + 0$ is zero. Both the successful and the failed comparison therefore return exactly $(0, w, 0, 0)$, regardless of whether $I(t, 0) = h$. Hence one command implements *Discard*, even with an arbitrary incoming cache. No disjointness assumption about the truth and implication codes is needed.

Omitting a successful-MP zero push was explored previously at larger state counts, including in the PA and ZFC material. No priority is claimed for that general idea. The new result is its complete 194-state realization, its ten-entry separate-head command compiler and exact physical binding, and its verified novelty against the current supplied frontier.

3 A complete ten-entry command compiler

3.1 Exact numerical operations and command contracts

Let

$$p(a,b)=\frac{(a+b)(a+b+1)}{2}+b, \qquad [a;w]=p(a,w), \qquad I(a,b)=4p(a,b), \qquad U(i,a)=4p(a,i)+2.$$

Iterated brackets put their leftmost entry at the top. The Cantor projections are total and satisfy

$$\mathrm{hd}[a;w]=a, \quad \mathrm{tl}[a;w]=w, \quad [0;0]=0, \quad \mathrm{hd}(0)=\mathrm{tl}(0)=0.$$

A numerical stack is therefore not being treated as a free list. A command store is (t,w,d,h): current formula, stack, offset, and cached formula. Code zero is truth and code one is the target contradiction. The new MP command, with $y=d+x$, is

$$\mathsf{MP}_x(t,w,d,h)=\begin{cases}(y,w,y,0), & I(t,y)=h,\\ (0,w,y,0), & I(t,y)\neq h.\end{cases} \tag{1}$$

Shift is retained literally:

$$S(t,w,d,h)=\begin{cases}\mathsf{halt}, & t+h=1,\\ (h,\mathrm{tl}(w),d,\mathrm{hd}(w)), & t+h\neq 1.\end{cases} \tag{2}$$

For $D=d+x$, $H=h+y$, the unchanged AX command has output

$$\mathsf{AX}_{A,x,y}(t,w,d,h)=\big(g(H,D,A),[q_1,\dots,q_9,t;w],0,0\big). \tag{3}$$

Here $q_1=\Gamma(A)=I(A,U(A,A))$, $q_2,\dots,q_7$ are the six retained PA196 heads in reverse emission order,

$$q_8=\rho(D,A,H,d)=I(I(I(H,d),D),I(I(D,H),I(A,H))), \qquad q_9=U(d,h).$$

The gate g is the original separate-head guarded separation gate. The free-constructor computation in `check_emitter_and_words.py` binds all these formula trees to the delivered grammar. All $q_1,\dots,q_8,g$ are divisible by four, and $q_9\equiv 2 \pmod 4$, with $q_9\geq 2$. In particular none is code one. This is a statement about constructor values, requiring no consistency assumption and holding for arbitrary incoming natural stores.

The count in (3) is nine generated stack entries plus the saved old top. The two identities needed below are

$$\mathsf{AX}_{A,x,y};S^9(t,w,d,h)=(q_8,[t;w],0,U(d,h)), \tag{4}$$

$$\mathsf{AX}_{A,x,y};S^{10}(t,w,d,h)=(U(d,h),w,0,t). \tag{5}$$

All displayed Shifts are nonhalting: their tests involve two generated non-one codes, zero and a generated code, or the universal code of value at least two. This also establishes exact stack effects when zero pairs collapse.

3.2 Uniform stack operations and a one-command discard

Write $A_0 = \mathsf{AX}_{0,0,0}$ and define

$$\mathsf{Push} = A_0; S^9; \mathsf{MP}_0, \qquad \mathsf{Pop} = A_0; S^{11}, \qquad \mathsf{Discard} = \mathsf{MP}_0, \qquad \mathsf{Offset}_i = \mathsf{MP}_i.$$

Their exact contracts are

$$\mathsf{Push}(t, w, d, h) = (0, [t; w], 0, 0), \tag{6}$$
$$\mathsf{Pop}(t, [a; w], d, h) = (t, w, 0, a), \tag{7}$$
$$\mathsf{Discard}(t, w, 0, h) = (0, w, 0, 0), \tag{8}$$
$$\mathsf{Offset}_i(0, w, 0, 0) = (0, w, i, 0). \tag{9}$$

For Push, (4) makes the final MP compare an implication with a universal code, so it fails with offset zero. For Pop, the first ten Shifts give (5). The eleventh Shift tests $U(d, h) + t \geq 2$, hence it does not halt even when $t = 1$. It pops exactly the displayed item. Thus Pop is total on all natural stores. In particular $\mathsf{Pop}(t, 0, d, h) = (t, 0, 0, 0)$ and $\mathsf{Push}(0, 0, d, h) = (0, 0, 0, 0)$.

For Discard, the two MP outcomes in (1) coincide when $d = x = 0$, for arbitrary h: both are $(0, w, 0, 0)$. This gives a single-command discard, where the old dummy-push controller needed a longer word. The old cache need not be zero for this identity. For Offset, if $i > 0$ then $I(0, i) > 0$, so the comparison fails; if $i = 0$ it succeeds and returns $(0, w, 0, 0)$. The same one-command definition covers both cases. These simplifications affect the proof-word compiler; they do not claim additional physical state savings.

3.3 Axiom selectors and generalization

In the following contracts the arbitrary incoming store is $(t, w, 0, 0)$. For $1 \leq j \leq 7$, the word

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; S^{j+1}; \mathsf{Pop}^{9-j}$$

returns $(q_j, w, 0, 0)$ with the parameters evaluated at the desired A, D, H. After the initial Shifts, the remaining $9 - j$ stack entries are exactly the unselected generated tail and the saved zero. The Pops preserve the selected top and consume those entries; the last pop puts that saved zero into the cache. For the dominated head, if $u \leq a$, use

$$\mathsf{Discard}; \mathsf{Offset}_u; \mathsf{AX}_{b,a-u,c}; S^9; \mathsf{Pop}.$$

It returns $(\rho(a, b, c, u), w, 0, 0)$, with the numerical dominance condition coming from $a = u + (a - u)$. A permitted gate is obtained by

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; \mathsf{Pop}^{10},$$

which returns $(g(H, D, A), w, 0, 0)$. Truth uses Discard alone.

The word

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{Offset}_i; S; \mathsf{AX}_{i,0,0}; S^{10}$$

returns $(U(i, a), w, 0, 0)$ from $(a, w, 0, 0)$. Before its AX the store is $(0, w, i, a)$; its last ten Shifts use (5) with saved top zero. Its first Shift tests $0 + 0$, and the later ones are among the safe AX shifts. Hence this generalization word is total even for $a = 1$.

3.4 Modus ponens with its spare zero supplied by the compiler

Suppose recursively compiled proofs return a and $I(a,b)$ with the stack framed and both caches cleared, or halt earlier. The MP compiler is

$$\mathsf{Discard};\mathsf{Push};C(I(a,b));\mathsf{Push};C(a);\mathsf{Push};S;S;\mathsf{MP}_b;\mathsf{Pop}. \tag{10}$$

The first two commands reserve a zero below the premise computations. If the recursive calls have returned, the store before the two Shifts is

$$(0,[a,I(a,b),0;w],0,0).$$

The first Shift returns $(0,[I(a,b),0;w],0,a)$. The second halts if $a=1$; otherwise it returns $(a,[0;w],0,I(a,b))$. MP succeeds and returns

$$(b,[0;w],b,0).$$

By the total Pop contract, the final word returns $(b,w,0,0)$, including $b=1$. The stack zero is now consumed once, at the compiler level. At $a=b=w=0$ the same proof uses the literal Cantor projection identities; no extra distinction between a zero push and an empty stack is introduced.

Induction on a finite derivation p in the inherited separate-head calculus therefore proves:

> From every $(t,w,0,0)$, $C(p)$ either halts or returns $(\mathrm{concl}(p),w,0,0)$.

The axiom and generalization cases use the words above. The MP case is (10); extending a word after an early halt still halts. In particular a refutation compiler ends in top one, if it has not already halted. Its final appended Shift halts because $1+0=1$. For arbitrary incoming t,w,d,h, first apply A_0 to clear d,h. Thus the concrete word

$$A_0;C(p);S$$

halts from *every* incoming store whenever p derives code one. The compiler is a primitive-recursive traversal of the finite derivation, and PA proves its correctness by induction on that coded derivation. It does not assume a reset between the scheduler's unsuccessful trials.

The unchanged scheduler enumerates finite command words with the usual Cantor continuation coding. A cofinal family of halting fresh candidates is obtained by prefixing the universal word by $\mathsf{AX}_{k,0,0}$. Let $E(\epsilon)=0$ and let a positive continuation be encoded by $E(c::w)=1+\mathrm{encodeCommand}(c,E(w))$. In particular $\mathrm{encodeCommand}(\mathsf{AX}_{A,x,y},z)=p(A+2,p(y,p(x,z)))$. Put $z=E(A_0;C(p);S)$, using this positive continuation convention rather than a fresh-candidate code. The fresh candidate codes for the prefixed words are exactly $p(k+2,p(0,p(0,z)))$, which are strictly increasing and unbounded in k. The fresh counter begins at two, so omitting smaller candidates loses none of this cofinal family. Every decoded continuation strictly decreases, and the unchanged source macro loops terminate. The resulting finite-word and finite-trace construction is formalizable in PA by primitive recursion and induction.

3.5 Soundness and the PA calibration

Annotate the current top t, cached head h, and every live canonical Cantor-stack entry by a derivation in the separate-head calculus, or by a PA proof of its arithmetic translation. For $w>0$, its decoded tail is strictly smaller than w; thus the canonical stack and its annotation have a primitive-recursive

bounded construction. A zero stack supplies truth when popped, and pushing zero onto zero preserves the empty canonical annotation.

AX adds the unchanged valid axiom instances, a generalization of the old cached theorem, and the saved old top. On a successful MP, the annotations of t and $I(t, y) = h$ produce an annotation for y. The stack is retained, and the cache receives the truth proof. Failure outputs truth and retains the stack. Shift transports the annotations. If its test succeeds, then $t + h = 1$ implies that one of the two annotated formulas is code one. The inherited syntactic arithmetic translation turns that annotation into a PA refutation. No annotation is imposed on the arithmetic offset d.

Consequently PA proves both finite syntactic reductions:

$$\text{source halt} \longrightarrow \text{PA refutation}, \qquad \text{PA refutation} \longrightarrow \text{source halt}.$$

For the second reduction, compose the supplied primitive-recursive PA-to-separated-head proof translation with the all-store compiler above. For the first, compose the proof annotation with the supplied separated-head-to-PA translation. This uses the existing logical calibration, including its treatment of the code-zero alias and coincident variable labels. Nothing in the no-push modification changes that calculus or requires a new logical axiom, a reflection principle, or consistency as an additional assumption.

4 The inherited logical calibration and PA internalization

The logical source is the same separate-diagonal-head numerical calculus as the principal PA194. Its constructors are

$$I(A, B) = 4P(A, B), \quad M(a, b) = 4P(a, b) + 1, \quad U(a, B) = 4P(B, a) + 2, \quad E(a, b) = 4P(a, b) + 3,$$

with negation $N(B) = I(B, 1)$. It retains the six quantified/equality heads, the numerically dominated propositional schema, theorem generalization, and modus ponens. The separate diagonal head is $\Gamma(B) = B \to U(B, B)$; the guarded pointed-separation gate is unchanged. AX saves ten entries, including the incoming top; the compiler in the preceding section uses this exact count.

The supplied calibration gives primitive-recursive transformations of finite refutations in both directions, with correctness provable in PA. It is a syntactic theorem about proof codes. The soundness direction interprets membership by finite binary membership, equality by arithmetic equality, and quantification by first-order quantification. For each fixed arithmetic formula θ, PA proves finite binary comprehension by induction on a finite bound. Substituting the syntax of θ into that proof template is primitive recursive; no uniform arithmetic truth predicate or PA reflection principle is asserted.

The converse derives the interfaces used by the original PA refutation compiler: both membership rewrites, neighborhood equality, and equality witnesses. Numerical code descent supplies the freshness of label B in its own decoded formula, so the diagonal head replaces the older stronger equality-vacuity interface. Bundled and separated diagonal gates have primitive-recursive proof translations. These are the logical reductions used here, not a substitution of iPRA consistency for PA consistency.

The supplied argument explicitly treats $I(0, 0) = 0$, $[0; 0] = 0$, and coincident variable labels in pointed separation. In particular, when the two bound labels coincide, the comprehension matrix is adjusted to use the singleton condition; treating the original free parameter as unchanged under that binder would be incorrect. The relevant full supplied calibration fragments are retained under `references/logical/`, with their attribution and exact source snapshot.

Let R_N be the modified finite register program and let H_R express a finite source halt from the verified clean seed. Command soundness labels every live formula and canonical stack cell with a finite PA

proof of its translation. The new successful MP removes a push of truth and leaves the previously annotated stack unchanged. The failed branch creates truth; Shift, AX, and scheduling retain their old annotation rules. If the sum of the two visible formula codes is one, one of the annotations is a proof of the contradiction code. This yields a PA refutation.

Conversely, the inherited PA-to-native proof translation followed by the new strong command compiler produces a finite halting command word. Its arbitrary-store contract and cofinal word encoding handle the scheduler's carried state and any finite initialization prefix. All these are primitive-recursive finite proof/trace constructions whose correctness is proved by ordinary induction in PA. Therefore

$$\mathrm{PA} \vdash H_R \leftrightarrow \exists p \, \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner).$$

The next section transports this statement to the literal 194-state table.

5 Physical construction and finite-trace preservation

The numerical axiom source, the physical register arrangement, and the seven simultaneous AX folds are unchanged. The state accounting is

$$\underbrace{134}_{\text{fixed controls}} + \underbrace{67}_{\text{raw AX readers}} - \underbrace{7}_{\text{compatible folds}} = 194.$$

The raw table has 201 controls. The seven folds are reconstructed, and all 28 context/read-bit contracts are replayed in the final simultaneous quotient. The check includes incoming transitions establishing the history bits; the altered MP row does not target a fold member. Full primitive source simulation checks the resulting table independently of that local quotient argument.

The logical register order is $(N, T, W, A, D, H, X, Y, L)$, with physical map $(6, 2, 0, 4, 3, 7, 8, 5, 1)$. A block $0\,1^{v+1}$ encodes natural value v. The source frame contains the left marker, the source-path bits, an affine zero gap, the delimiter, the nine blocks, and an arbitrary finite nonempty list of spare blocks. The header distance is $48 + n$, for arbitrary $n \geq 0$. The increment operation's append-one spare effect is included.

Exact blank execution reaches source site 25, native state 84, with head -2 and logical registers $(2, 0, 0, 0, 0, 0, 0, 0, 0)$, after 58,154 transitions. The tape check compares its entire finite support, every source bit and marker, all nine register blocks, all 26 one-valued spare blocks, and zero exterior. The source gap invariant starts at this exact frame.

Lemma 5.1 (Positive primitive simulation)**.** *Each nonhalting source instruction of R_N has a positive, terminating literal-tape realization from every permitted source frame. It reaches exactly the prescribed successor frame with the specified register and spare-tail updates. Both branches of every decrement are covered. A source halt reaches the physical terminal control.*

Proof. The delivered finite instruction map has 3,654 sites and 4,755 guarded outcomes. The construction checker verifies every one using the fixed register-engine contracts. The independent checker derives four generic list-sweep lemmas from the actual transition rows and checks the same outcomes using a different literal-tape representation and independently resolved control flow. Its proof language consists of literal steps, constant-symbol scans, and generic finite-list sweeps. Induction on unary block length and spare-list length justifies those rules. Instantiating the finite derivations proves the contracts for arbitrary frame parameters. Every segment has positive duration; the smallest certified affine constant is three. These elementary induction arguments and finite certificate instances are formalizable in PA. □

Theorem 5.2 (PA-provable source/tape halting equivalence)**.**

$$\mathrm{PA} \vdash H_N \leftrightarrow H_R.$$

Proof. A halting source trace gives a halting tape trace by concatenating its primitive simulations after the exact startup. Conversely, if the tape first halts after T steps, iterate the total source step function for $T + 1$ instructions from the clean seed, making a source halt absorbing. If no source halt occurs, positive primitive simulations concatenate to a nonhalting tape prefix of length greater than T, contradicting machine determinism. All iterations and finite trace concatenations are primitive recursive. PA proves their correctness by induction; the argument does not assume the actual run is nonhalting. □

Composing with the preceding logical/source equivalence proves the theorem in Section 1.

6 Verification results and exact scope

Check	Observed result
Working controls / literal rows	194 / 388
Raw AX readers / compatible folds	67 / 7
Primitive source sites / outcomes	3,654 / 4,755
Construction-checker symbolic steps	1,473,292
Independent-checker symbolic steps	1,073,769
Independent maximum per outcome	397
Independent minimum duration constant	3
Generic loop lemmas rederived from rows	4
Exact complete-tape blank startup	58,154 steps
Previously represented 194-state graph classes	8
Previously represented 194-state byte forms	15

Symbolic-step totals measure verifier work, not a uniform literal running time for arbitrary natural register inputs. Startup is a finite exact initialization check; it is not empirical evidence for infinite nonhalting. The universal contracts cover nine independent natural register values, an arbitrary affine gap and an arbitrary nonempty finite spare list. The two checkers are independently implemented at the physical level; they are not two independently developed logical PA calibrations.

The corpus novelty checker compares exact canonical transition encodings, not only differing file hashes. The new root-free graph hash is

`e74434eb401367ab63d712bbde26a0edcec17b2ac536ca4d474e743d9969b036`.

It differs from every recognized existing 194-state graph even after arbitrary state renaming, changing a graph-reaching root, tape reflection, and simultaneous read/write symbol complementation. The last symmetry is excluded only as cosmetic novelty; it would alter the specified blank symbol if used as a machine transformation.

The mathematical statement is an ordinary PA-provability theorem built from the explicit proof transformations and physical induction arguments. The field `kernel_checked:false` records that no new end-to-end Lean/Rocq theorem was compiled. The repository's open machine-specific iPRA obligations remain separate. No Python run alone proves the inherited logical calibration.

7 Comparison with the complete supplied corpus

The comparison is pinned to ProveIt commit `17f92105e2bc4317feae6c219cb2143437b482ff`. At this revision the PA research directory contains the extracted contents of 36 delivered packets. The incoming directory contains 26 ZIP files: 21 are retained copies of those packets, and five are later cached-controller packets. Thus there are 41 top-level research packets to compare, rather than 62 independent constructions. The numerous `pa194-*` and `pa195-*` proof directories develop different proof layers for the same designated machines; their names do not signify additional machine designs.

The comparison has two levels. First, a fresh read-only census scans every `.tm` and `.txt` file under the entire PA directory, every incoming ZIP, and every ZIP recursively encountered in those sources. Archive payloads are deduplicated by their complete SHA-256. Second, the reports and source descriptions classify the recognized tables as principal results, supplementary implementations, search seeds, or rejected experiments. A small state count or a filename beginning with "candidate" is never sufficient for a positive classification.

The census visits 44 distinct ZIP payloads through 71 archive references. It recognizes 4,417 table occurrences, 620 exact byte identities, and 446 action-preserving, root-free control-graph classes. Two disconnected synthetic controls have no root reaching their whole graph and are left unclassified. The numeric tables contain eight 194-state graph classes, sixteen 195-state classes, and ninety-nine 196-state classes. These are corpus counts, including diagnostics and repeats of ancestors; they are not counts of established consistency machines. No recognized complete table has between 10 and 193 working states. This observation is not a state-count lower bound.

7.1 The eight existing 194-state graphs

Family	Controls / source sites	Change and status
PA194-67	134 + 67 - 7; 3673 sites	Separate diagonal head; 67 AX readers and seven compatible history folds. principal; ordinary PA proof and two symbolic physical checkers
PA194-AD	134 + 67 - 7; 3673 sites	Exchange the physical A and D register positions; three changed native rows. principal companion; ordinary transfer and two symbolic physical checkers
PA194-68	134 + 68 - 8; 3673 sites	Different association of the same separate-head emitter, with eight folds. principal; ordinary PA proof with independently reconstructed source/table
PA194C	134 + 67 - 7; 3649 sites	Cached Shift/MP controller on PA194-67; six changed native rows. All five later cached packets duplicate this exact dense table. principal; ordinary inherited PA calibration and two symbolic physical checkers
PA194CS	135 + 67 - 7 - 1; 3646 sites	Destructive equality operand symmetry shortens Shift by three sites. Extra controller reader 330 is folded with 41; this is not an extra AX fold. principal; source block equivalence and two symbolic physical checkers
PA194altC	134 + 68 - 8; 3649 sites	Cached controller composed with the alternate PA194-68 reader allocation. search seed; no separate complete consistency endpoint delivered

Family	Controls / source sites	Change and status
PA194CZ	134 + 67 - 7; 3649 sites	Replace source-site-26 jump by a zero test of register six; native row (27,1) changes from 0 L 40 to 1 R 21. supplementary seed; primary contracts and positivity-mask invariant, no new independent physical replay
Rejected scanner194	rejected	Optimistic shortened-scanner/reader quotient fails the required source simulation. explicitly rejected frame-contract experiment; not a Con(PA) result

The five principal designs in the table are PA194-67, PA194-AD, PA194-68, PA194C, and PA194CS. The remaining three are deliberately retained in the novelty comparison: PA194altC and PA194CZ already occur as supplementary search seeds, while the shortened-scanner table is explicitly rejected. All eight classes remain distinct even if one allows tape reflection, simultaneous read/write symbol complementation, and arbitrary choice of a graph-reaching root. Symbol complementation is used only as a stronger test against cosmetic novelty; it does not preserve the specified all-zero blank input.

Several archive names suggest different cached machines, but their principal literal result is the same table. Its SHA-256 is

`d759313c03b515293fe06b12769fa1771d3829a2bad89febaa9b4391990f3a4b`.

In particular, recomposing the primary seven-fold decoder with the Combined controller would reproduce an existing result. The symmetry of the destructive equality test is also already used by PA194CS. Likewise, merely installing the Combined controller on the alternate eight-fold decoder, or adding the zero-register jump at source site 26, would reproduce supplied search seeds. A proposed fallback must be compared with these supplementary tables as well as the principal ones.

7.2 The positive 195-state families and their limitations

Family	Sites	Outcomes	Startup	Physical organization
Original PA195	3,700	4,815	58,154	$134 + 68 - 7$
195S	3,673	4,780	58,154	$134 + 67 - 6$
195C	3,649	4,749	57,922	195S with Combined commands
M195S8	3,700	4,815	59,164	Extra scanner control and eight folds
PA195DC	3,676	4,784	57,922	Original diagonal emitter, Combined commands

The table identifies the five designated positive manuscript families. Original PA195 packages the diagonal-vacuity formula with the guarded gate. The separate-head emitter saves one macro occurrence and 27 primitive sites; it underlies 195S, 195C, and the principal 194-state families. Combined Shift and MP remove a further 24 primitive sites, but do not automatically save a reader state. M195S8 uses an additional scanner control to make a different eight-fold allocation valid. PA195DC applies the Combined controller to the original bundled-diagonal source rather than the separate-head source.

The sixteen recognized 195-state graph classes include these five, six explicitly rejected or diagnostic quotients, a secondary separate195 layout, a crossed allocation, the 195-state parent of PA194CS, and two secondary implementations in the PA195DC search material. The latter include a negative-diagonal variant whose author explicitly keeps it outside the principal consistency theorem. They must not be silently promoted to independently proved consistency constructions. The full class inventory records every byte identity and source locator.

7.3 What the larger designs contribute

James Chang's PA209 and the Joint/Compatibility U209 ancestor are different 209-state constructions, with respectively 3,948 and 3,909 primitive source sites. The U208 packets identify two old decoder roles by a finite-history argument. Entry-context, three-cell, and history presentations repeatedly describe this same kind of one-state saving; later helper layouts produce further 208-state graphs. Their equal state counts do not identify their source programs or transition graphs.

The published PA196 has 3,812 source sites and 4,959 primitive outcomes. Its allocation comprises 36 tape/register-engine controls, 13 primitive routers, and 147 readers. The later 196-state proposals change head ordering, association, physical A/D positions, closing decrements, or zero-register jump routing. The direct repository catalogue selects 22 rooted 196-state graph classes and explicitly distinguishes two experimental MP variants and two rejected layouts. The latter both halt on blank tape after 43,563 transitions; their finite halting traces are Lean checked in the repository. They are controls, not consistency machines.

The mathematical source changes show why source length cannot be used as a proxy for the number of physical states. Diagonal vacuity reduces the emitter to 3,700 sites, but its first implementation uses 199 states; later allocations reach 198, 196, and 195. Separating the diagonal head gives 3,673 sites and eventually reaches 194. An integrated witness gate uses 3,745 sites and 4,872 primitive outcomes, yet its verified physical allocations use 206 and then 203 states. Distributed-gate and short-negation alternatives likewise lose advantageous reader sharing. The NEG18 separate-emitter source has only 3,441 sites and 4,468 outcomes, but the documented allocation is $134 + 68 - 5 = 197$. A repaired 157-node representation of the unchanged PA196 source has a 207-state realization: a smaller abstract DAG can require more routing and fewer useful contextual identifications.

Changed command protocols form a separate axis. H-only Shift alone has 3,811 sites; Combined196 has 3,788 sites, 4,928 outcomes, and a complete 57,922-step startup. Its successful MP returns the consequent in H, clears T and D, and does not push a dummy stack value. Shift tests H. These semantics differ from the original controller, so the command compiler and arbitrary-incoming-store refutation argument must change with them. An MP-only bounded register experiment is not the same result as a complete Combined implementation with all universal source outcomes.

Other packets contain source-only grammar proposals, failed scanner shortcuts, pruned sources without a calibrated blank seed, and intermediate parity/control components. Their contribution is evidence about particular search spaces or reusable proof obligations. A failed universal frame contract rejects the proposed simulation argument; it does not by itself prove that the failing frame occurs on the blank run. Conversely, a long nonhalting prefix does not repair that argument.

7.4 Proof status and inherited assumptions

The ordinary PA calibration in the recent reports is stronger than experimental evidence but distinct from a freshly elaborated proof-assistant endpoint. It supplies syntactic translations of PA and numerical-calculus refutations, an explicit command compiler, exhaustive command enumeration, and positive-duration physical simulations. The reports explicitly address the code-zero alias $I(0,0) = 0$, the Cantor-stack alias $[0;0] = 0$, early halts during proof compilation, an arbitrary carried store between trials, and the coincident-variable case in the finite bit-comprehension argument. These points matter to the claimed PA-provable biconditional.

The repository's current README nevertheless states that neither machine-specific reduction has yet been established in iPRA for the primary PA194, original PA195, or PA196 candidate. Its ordinary finite-run Lean theorems and conditional arithmetic endpoints do not close those obligations. This is

consistent with reusing an explicit ordinary mathematical PA calibration; it must not be described as a completed Lean/Rocq or iPRA result. The present comparison audits the arguments and their interfaces but does not independently rebuild every historical proof-assistant development.

7.5 Coverage limits and reproducibility

The literal recognizer requires a UTF-8 text file of five-column rows `state read write direction next`, numeric state labels, a binary alphabet, both rows for every working state, and exactly one undefined halt target. Blank lines and whole-line comments are allowed. Files using other syntaxes, multiple undefined halts, code-generated tables not materialized as text, and tables embedded solely in JSON or Lean are outside this mechanical recognizer. They are considered through the reports and source descriptions rather than counted as recognized literal tables. The archive ledger gives all 41 top-level packet contributions.

Graph classification forgets the designated start and compares each root that reaches the full working graph by deterministic breadth-first numbering with symbol-labelled edges and exact write/direction actions. Taking the least such encoding yields a complete invariant for this stated class of root-free graphs. This is stronger than comparing native labels or file hashes, but makes no assertion about semantic equivalence or correctness. Exact machine bytes, graph hashes, all occurrence paths, archive identities, the parser, and the read-only reproduction command are included in the comparison artifacts.

7.6 Prior work on expanded macro searches

The supplied search material already contains a partial expansion of the parameter-copy macros. In `pa194_conpa_2026-11-13_2/search/exploratory/expanded_input.txt`, fourteen terminals replace the eleven-terminal basis. Native copy roots 12, 98, and 166 are opened into push root 19 and tail roots 20, 114, and 179. The old diagonal macro word grows from 191 to 252 terminals. The associated `search_expanded.cpp` hashes newly composed pairs against all fixed pure-reader pairs, so exact fixed-pair reuse is already present. The recorded expanded and general runs cover 3,363,000 and 1,023,000 iteration slots, respectively, and retain 195 states as their best count. Earlier free-macro-aware grammar searches also give source-only upper bounds.

Opening deeper interiors of the cons and push macros, using the newer separate-head or cached source, and performing a physically checked global optimization can extend those searches. It would be inaccurate to claim that exposing the three copy tails or recognizing fixed pure reader pairs is itself a new idea. Nor do those negative runs prove that a deeper or differently initialized search cannot save a state.

8 Search below 194 and its limits

The positive deliverable is PA194N. No smaller machine was certified. The new search effort addressed two concrete limitations of the supplied searches: treating large fixed subroutines as indivisible leaves, and requiring every incoming edge of an eliminated state to share one replacement. It also re-examined an outer-controller rotation motivated by a possible shorter return scanner.

8.1 Deeper macro expansion and broader fold allocation

The first decoder experiment opens one pure-reader level of every eligible axiom macro, including the Cantor push and constructor routines as well as the three parameter copies. A deeper experiment opens a second level of the push, constructor and guard routines while retaining the copy-tail interfaces. These extend the earlier three-copy-tail experiment identified in the corpus review. The separate-head source and its new no-push controller supply different starting points from that earlier diagonal source.

The search interns new binary reader pairs against the fixed controller's existing pure-reader pairs. It uses both whole-shared-subtree and individual occurrence rotations. In broader runs, a candidate fold may identify an AX reader with a fixed control, provided all incoming rows establish the required preceding bit. The local simulation is checked in the simultaneously patched table. One heuristic variant prunes partial fold selections that already fail their local check; that is a search restriction, not a completeness theorem about possible quotients.

Search	Seconds	Proposal slots	Best score
One-level expansion; AX folds	600	6,548,653	194
One-level expansion; fixed/AX folds	600	3,584,338	194
Partial-selection pruning	600	3,487,411	194
No-push controller; one level	700	5,228,572	194
No-push controller; deeper interiors	700	6,366,489	194
Total	3,200	25,215,463	194

An iteration is a mutation-proposal slot; it need not denote a distinct candidate, and some slots produce no mutation. The reported best value is only a screened allocation score, equal to 134 plus reader count minus compatible folds. No value below 194 was reached. A grammar achieving a promising score would still have to pass exact source-graph binding, complete physical simulation, and calibrated blank initialization. These heuristic searches therefore establish neither optimal grammar size nor a state-count lower bound. The accepted PA194N does not depend on their termination or on the optimality of their result.

The frozen inputs, source maps, search sources, random seeds, complete logs and terminal iteration totals are included. Time-limited runs can have different iteration counts on different computers; the retained data, not a promise of deterministic stochastic discovery, defines the evidence.

8.2 Additional physical contraction searches

Two further searches were carried out on the existing PA194CS table and on the new no-push table. The contexts were regenerated from all source outcomes, symbolic unary-sweep interiors, generic list-block proofs, and the exact blank initialization. Every window has radius five, and every local replacement has a 300-transition bound.

input	source outcomes	symbolic views	windows
PA194CS	4745	19067	29035
PA194N	4755	19043	29017

The supplied common-action merger yielded no additional state identification on either set. A new *incoming-edge bypass* search also yielded none. The latter does not insist on one common survivor for all incoming edges of a deleted control: it permits each incoming transition to receive its own new action, provided its bounded continuation reproduces the old incoming step followed by the deleted

control's step. The continuation must preserve the entire window, avoid the deleted control and all rewritten rows, and end at the exact old control and head position. Self-looping removed controls are excluded from this search rule.

These are bounded searches through sufficient rewrite rules. They are not lower bounds for binary Turing machines and do not exclude larger simultaneous rewrites or a different register engine. The frozen tables, context files, search sources and empty candidate outputs with progress logs are supplied in the supplementary search bundle.

A startup/scanner obstruction. Reassociating the outer source tree from $((G, D), C)$ to $(G, (D, C))$ removes one leading zero from the first comparison path. Its resolved source graph changes only the final relative continue target, which would need adjustment from three to four. However, the obvious four-row physical implementation is invalid: the shared native control 69 implements the decoder block only with a preceding one. The rotated entry supplies a preceding zero and selects its other shared behavior. The candidate fails the universal contract at source site 11, and its blank startup does not reach the intended clean frame. This is a rejected experiment, not a 193-state result.

8.3 What transfers from ZFC and concise iPRA

The ZFC work supplied concrete precedents for removing the successful-MP zero push, proving proof compilers over arbitrary retained stacks, and separating a source change from its physical state price. PA194N uses that interface discipline and keeps the PA-calibrated numerical calculus. Its one-command reset is an exact consequence of the altered MP branches; it is not an additional logical axiom.

The concise-iPRA project provides useful examples of deriving apparently necessary rule cases from a smaller interface: one-term transport, restricted-source transport, and combined introduction rules. That suggests examining special cases and interface redundancy before shortening code. It does not license replacing the PA consistency search by an iPRA consistency search. A new smaller axiom basis would require its own PA-verifiable proof translations and physical compilation. No such replacement was established here.

The largest remaining opportunities are a jointly changed fixed controller and AX layout, a new source interface with a smaller physical price, or a correct global return-scanner redesign. The outer-rotation failure shows why a syntactically shorter path cannot simply be substituted into a history-sensitive decoder. These are open directions, not promises of a 193-state construction.

9 Reproduction and package organization

Python 3.10 or later and the standard library suffice for the principal construction, command certificates, and independent physical check. Assertions must remain enabled. The root `README.md` gives exact commands, and `verify_all.py` runs the accepted checks from any current directory without network access. Search replay alone requires a C++17 compiler and is not necessary to reconstruct the accepted table. A standard TeX Live installation builds this report with two passes of `pdflatex` from the `report/` directory.

The archive separates accepted construction, independent verification, mathematical proofs, comparison, inherited references and exploratory search artifacts. Rejected or unproved search proposals are not used in the theorem. Exact bytes, snapshot identities and licensing notices are recorded with the artifacts. No repository modification or publication is part of this delivery.

A Complete transition table

An entry wDq writes bit w, moves in direction $D \in \{L, R\}$ and enters q. H denotes the separate halt label 194. The start is state zero on the all-zero tape. The appendix is generated from the literal `pa194N.tm` file.

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
0	1R1	1R2	65	0R91	1R49	130	0R134	1R135
1	1R3	1R1	66	0R92	0L75	131	0R136	1R137
2	0R0	1R2	67	0R86	1R33	132	0R138	1R139
3	0R4	0R1	68	0R93	1R68	133	0R140	1R141
4	1R5	1L6	69	0L94	1L33	134	0R100	1R67
5	1L7	1L8	70	0L84	0R2	135	0R142	1R127
6	0L7	1L6	71	0R95	1R71	136	0R143	1R144
7	0L9	1L6	72	0R75	1R96	137	0R145	1R146
8	0L10	0L11	73	0R59	0R85	138	0R134	1R147
9	1R12	0L9	74	0R86	1R97	139	0R120	1R148
10	0L9	0L10	75	0R42	0L13	140	0R149	1L131
11	1L8	1L11	76	0R87	1R37	141	0R150	1R151
12	0R12	0R13	77	0R30	1R98	142	0R79	1R41
13	0L14	0R13	78	0R42	1R99	143	0R28	1R152
14	1L15	0L15	79	0R100	1R101	144	0L153	1L154
15	0L15	1L16	80	0R102	1L103	145	0L129	1R155
16	0L17	1L16	81	0R30	1R80	146	0R156	1R157
17	0L18	1L16	82	0R97	1R104	147	0R149	1R158
18	0L19	1L16	83	0R30	1R105	148	0R159	1R120
19	0L20	1L16	84	1R84	1L15	149	0R100	1R160
20	0L21	1L16	85	0R106	1R85	150	0R161	1R162
21	0L22	1L16	86	0R73	1R107	151	0R163	0L10
22	0L23	1L16	87	0L46	0R93	152	0R164	1R165
23	0R23	1R24	88	0R108	1R98	153	0R166	1R157
24	0R25	1R26	89	0R108	1R69	154	0R167	1R156
25	0R27	1R28	90	0R109	0L14	155	0R135	1R130
26	0R29	1R30	91	0R110	1L74	156	0R147	1R41
27	0R31	1RH	92	0R111	1R112	157	0R134	1R168
28	0L32	1R33	93	0R113	1R93	158	0R41	1L169
29	0R34	1R35	94	0R50	1R103	159	0L170	1R171
30	0L36	1L37	95	0R114	1R95	160	0R91	1R64
31	0R38	1R39	96	0R30	1R49	161	0R172	1R170
32	0R40	1R41	97	0R48	1R61	162	0R173	1R174
33	0R42	1R43	98	0R87	1L115	163	0R175	1R176
34	0R44	1R45	99	0R104	0L14	164	0R177	1R178
35	0R46	1R47	100	0R36	1R116	165	0L179	1L180
36	0R48	1R49	101	0R115	1R78	166	0R149	1R140
37	0R50	1R51	102	0L46	0R113	167	0R165	1R181
38	0R52	1R42	103	0L45	1R117	168	0L130	1R120
39	0R53	1R54	104	0L94	1R118	169	0R182	1R135
40	0R55	1R56	105	0R119	0L14	170	0R183	1R145
41	0R57	1R58	106	0R71	1R106	171	0R158	1R130
42	0R59	0R60	107	0R120	1R30	172	0R139	1R165
43	0R61	0L14	108	0L46	0R106	173	0R158	1R184
44	0R62	1R63	109	0R70	0R95	174	0R185	1R91
45	0R55	1R64	110	0R59	0R95	175	0R31	1R53
46	0L65	1R66	111	0L65	1R121	176	0R186	1R175

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
47	0R36	1R67	112	0R122	1R123	177	0R134	1R142
48	0R59	0R68	113	0R14	1R113	178	0R32	1R187
49	0R69	0L14	114	0R68	1R114	179	0R144	1R188
50	0R70	0R71	115	0R124	1L67	180	0R189	1R190
51	0R72	0L49	116	0R125	1R126	181	0R41	1R191
52	0R73	1R54	117	0R120	0L14	182	0R168	1R41
53	0L74	0L15	118	0R94	1L98	183	0R192	1R120
54	0R75	0L10	119	0R127	1R127	184	0R153	1R146
55	0R76	1R77	120	0R70	0R68	185	0R165	1R180
56	0R78	1R79	121	0R128	1R129	186	0R96	1R115
57	0R80	1R81	122	0R32	1R36	187	0R193	1R159
58	0R82	1R83	123	0R31	1R101	188	0R169	1R154
59	0L84	0R1	124	0R59	0R114	189	0R158	1R192
60	0R85	1R60	125	0R102	1R117	190	0R179	1R182
61	0L76	1L86	126	0R82	1R81	191	0R182	1R41
62	0R87	0L7	127	0R120	1R120	192	0R149	1R168
63	0R88	1R89	128	0R130	1R131	193	0R79	1R144
64	0R42	1R90	129	0R132	1R133			

B Archive-by-archive ledger

Packet	Reported contribution
pa_unary_power_cover	Unary power covering at PA strength (report of 27 October 2026). The U208 packets carry it as a baseline.
U208_ConPA_Conditional_Context_2026-10-29	U208 by a conditional tape-context reduction. States a 208-state candidate machine/u208.tm.
U208_ConPA_Entry_Context_2026-10-29	U208 by eliminating a state using entry context. States a 208-state machine machine/u208.tm.
U208_ConPA_History_2026-10-29	U208 by a finite-history decoder fold. States a 208-state candidate machine/u208.tm.
U208_ConPA_2026-10-29	U208 by a three-cell fold. States a 208-state candidate machine/u208.tm.
u208_research_2026-10-29	U208 research release. States two 208-state machines, machine/u208.tm and machine/variant_plain/u208.tm.
Below_U208_Research_2026-10-29	Research toward fewer than 208 states. States that no machine with 207 states or fewer was obtained.
u208_comparison_research_2026-10-29	Comparison of the 208-state proposals and a search below 208. States that the target was not achieved.
ConPA_below208_research_2026-10-29	Comparative audit and reduction search below 208. States that it does not establish a 207-state machine.
pa_consistency_research_2026-10-29	Audit and new source reductions. States that no validated machine below 208 states was obtained.
pa208_research_2026-10-29	Comparison and research below 208 states. States that it does not establish a valid machine below 208.

Packet	Reported contribution
pa_tm_research_2026-11-13	Comparison and reduction research. States a new 196-state variant with shorter source paths and nothing below 196.
pa_consistency_research_2026-11-13	Comparison and diagonal-vacuity research. States a shorter axiom program with a 199-state implementation and nothing below 196.
conpa_tm_comparison_and_reduction_search	Comparison and reduction search. States two experimental 196-state command variants and nothing below 196.
pa196_research_2026-11-13	PA196 comparison, source factorization and proof obligations. States a source tree of 157 internal nodes and nothing below 196.
pa_consistency_research	Research and verification packet. States that no machine below 196 states has been validated.
pa_tm_comparison_and_reductions	Comparison and new rewrites, a research snapshot.
pa_tm_extended_research_2026-11-13	Comparison and refined reduction search. States that the smallest accepted count is 196.
conpa_tm_comparison_and_search	Comparison and a search below 196. States that none was obtained; files named `195` are rejected or diagnostic candidates.
pa_consistency_reduction	Comparison and reduction experiments. States two new 196-state variants and a shorter source program with a 203-state implementation.
pa_consistency_comparison_2026_11_13	Comparison and new reduction experiments. States that nothing below 196 was obtained.
PA_consistency_TM_research_2026-11-13	Comparison and new source reductions. States that no machine below 196 is claimed.
pa_consistency_research_2026-11-13_2	Comparison and search below 196. States two new layouts with 196 and 198 states and a command-level reduction.
conpa_tm_study_2026-11-13	Comparison and reduction study. States that nothing below 196 was obtained.
pa_below196_research	Comparison and reduction experiments. States that nothing below 196 was obtained.
pa_consistency_comparison_and_new196_2026-11-13	Comparison and reduction research. States that it contains no accepted machine below 196.
pa_tm_research_2026-11-13_2	Comparison and reduction investigation. States same-size work, failed candidates and nothing below 196.
pa_consistency_cross_archive	Cross-archive constructions and reduction search. States that it does not certify a 195-state machine.
pa195_conpa_2026-11-13	PA195. States a 195-state machine `pa195/pa195.tm` and an article with a PA proof of the equivalence.
pa_consistency_reduction_audit_2026-11-14	Comparison and reduction audit. States nothing below 196 and no new theorem.
pa_consistency_research_2026-11-14	Comparison and NEG18. States an 18-instruction encoded negation with a compiled machine of 197 states.

Packet	Reported contribution
`PA195_ConPA_research_2026-11-14`	A 195-state machine. States the primary `machine_195S.tm` and a second table `machine_195C.tm`.
`pa195_followup_research`	Comparison and searches below 195. States a graph-distinct 195-state machine M195S8 and nothing below 195.
`pa194_conpa_2026-11-13`	A 194-state machine. States the table `construction/tables/pa194.tm` and a proof of the equivalence in PA.
`pa194_conpa_2026-11-13_2`	PA194. States a 194-state machine `pa194/pa194.tm` and an article with a PA proof of the equivalence.
`pa195_diagonal_combined_2026-11-13`	PA195DC. States a different 195-state machine.
`pa194C_conpa_2026-11-13(2)`	Same PA194C literal table; independent whole-tape symbolic checks, cached command algebra, corpus comparison.
`pa194_new_designs_2026-11-13`	Same PA194C plus PA194CS symmetric-Shift implementation; 3646 source sites and additional controller fold.
`pa194c_conpa`	Same PA194C; refreshed comparison and secondary PA194altC 68-reader/eight-fold cached search seed.
`pa194c_conpa_2026-11-13(1)`	Same PA194C; supplemental PA194CZ zero-register-jump seed, not a state saving.
`pa194c_conpa_2026-11-13`	Same PA194C; inherited calibration, command checks and independent physical checker; 34-archive historical census.

C Independent review and artifact boundary

A separate mathematical review checked the command proof before packaging. It confirms the total Pop contract even at formula code one, the one-command Discard identity for arbitrary incoming cache, the exact ten-entry selectors, the reserved-zero MP compiler, and the arbitrary-incoming-store refutation word. It also checks the ordinary proof-annotation converse. The complete review is supplied as `mathematics/independent_review.tex`; it is not represented as an external human endorsement or a proof-assistant check.

The independent physical review uses different tape representations and control-flow resolvers, and no compiler or arithmetic-box oracle. Its portable source and full receipts are under `verification/independent/`. The logical calibration, source-level compiler and physical semantics remain explicit, separate dependencies in the manifest.

Sources and attribution

1. Vladimir Reshetnikov, *ProveIt*, snapshot `17f92105e2bc4317feae6c219cb2143437b482ff`. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Computability/BusyBeaver/Research/PA. The comparison ledger identifies the complete packet corpus.
2. Primary separate-head PA194 packet: `PA/docs/pa194_conpa_2026-11-13/`, including the 67-reader/seven-fold decoder, source map, ordinary calibration and two physical checkers. The unchanged source fragments and licenses are retained in this archive.
3. Earlier PA command-reduction research, with the no-successful-MP zero-push interface, and the original ten-entry separate-head calibration. Exact file identities are recorded in the construction provenance.
4. James Chang's PA196 and PA209 developments, imported under `PA/jamesyc-upstream/`; the retained register compiler, numerical coding and source-language libraries are credited to their supplied notices.
5. ZFC research, especially `candidate-k215/alternatives/zero-spare-transfer/report.tex`, its no-successful-MP-zero-push discussion, and the source/physical preservation methods. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Computability/BusyBeaver/Research/ZFC.
6. ConciseIPRA native-core presentations and their exact interface reductions. https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA. No equivalence between Con(iPRA) and Con(PA) is assumed.